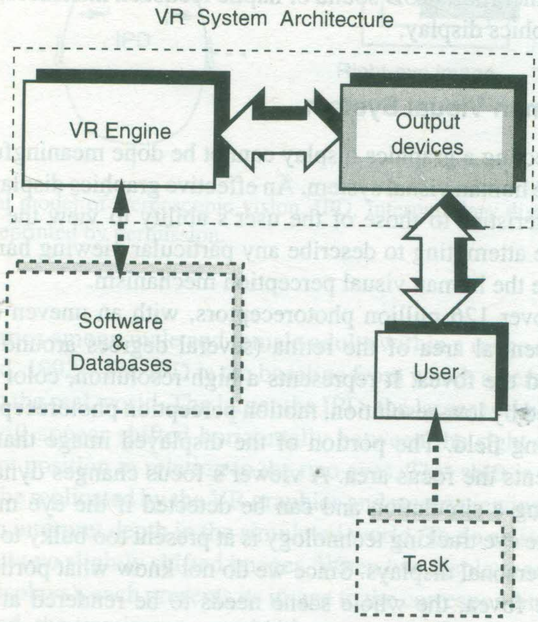
CHAPTER 3

OUTPUT DEVICES: GRAPHICS, THREE-DIMENSIONAL SOUND, AND HAPTIC DISPLAYS



The previous chapter described 3D trackers, trackballs, and sensing gloves, which are devices used to mediate the user's input into the VR simulation. Now we look at special hardware designed to provide feedback from the simulation in response to this input. The sensorial channels fed back by these interfaces are sight (through graphics displays), sound (through 3D sound displays), and touch (through haptic displays).

For reasons of simplicity, this chapter treats each sensorial feedback modality separately. A further simplification here is the decoupling of sensorial feedback from the user's input. Modern VR systems are, however, multimodal, and feedback interfaces usually incorporate hardware to allow user input (such as trackers, pushbuttons, etc.). Combining several types of sensorial feedback in a simulation increases the user's immersion into VR. Furthermore, combining several sensorial modalities improves the simulation realism and therefore the usefulness of VR applications.

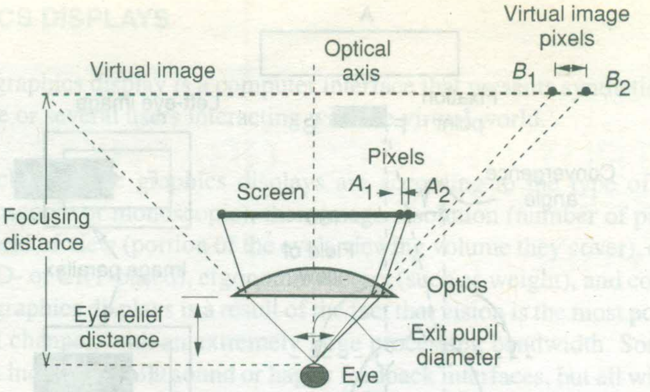


Fig. 3.2 Simplified optics model of an HMD. Adapted from Robinett and Rolland [1992]. ©1992 Massachusetts Institute of Technology. Reprinted by permission.

3.1.2 Personal Graphics Displays

Definition A graphics display that outputs a virtual scene destined to be viewed by a single user is called a personal graphics display. Such images may be monoscopic or stereoscopic, monocular (for a single eye), or binocular (displayed on both eyes).

Within the personal display category are grouped head-mounted displays (HMDs), hand-supported displays, floor-supported displays, and autostereoscopic monitors.

3.1.2.1 Head-Mounted Displays. These project an image floating some 1–5 m (3–15 ft) in front of the user [Anon, 2001], as illustrated in Figure 3.2. They use special optics placed between the HMD small image panels and the user's eyes in order to allow the eyes to focus at such short distances without tiring easily. Optics is also needed to magnify the small panel image to fill as much as possible of the eyes' field of view (FOV). As can be seen in Figure 3.2 [Robinett and Rolland, 1992], the unfortunate byproduct is that the distance between display pixels (A_1, A_2) is amplified as well. Therefore the "granularity" of the HMD displays (expressed in arc-minutes/pixel) becomes apparent in the virtual image. The lower the HMD resolution and the higher the FOV, the greater is the number of arc-minutes of eye view corresponding to each pixel. Older HMDs had very low resolution (as discussed in Chapter 1) and incorporated a diffuser sheet overlaid on the input to their optics. These diffusers blurred the edges between pixels somewhat, and the image was perceived to be better. Modern HMDs have resolutions up to extended video graphics array (XVGA; 1024 × 768) or better. Even these resolutions may be unsatisfactory if the optics amplifies the pixel dimension too much in order to enlarge the apparent FOV. Furthermore, very large FOVs force large exit pupil diameters, which can produce shadows around the edges of the perceived image.

Another HMD characteristic is the display technology used. Consumer-grade HMDs use LCD displays, while more expensive professional-grade devices incorporate CRT-based displays, which tend to have higher resolution. Since consumer HMDs are designed primarily for private viewing of TV programs and for video games rather than for VR, they accept NTSC (in Europe, phase alternating line, PAL) monoscopic video input. When integrated within a VR system, they require the conversion of the graphics pipeline output signal red–green–blue (RGB) format to NTSC/PAL, as illustrated in Figure 3.3*a*. The HMD controller allows manual adjustment for brightness and forwards the same signal to both HMD displays. Professional-grade HMDs are designed specifically for VR interaction and accept RGB video input. As illustrated in Figure 3.3*b*, in the graphics pipeline two RGB signals are sent directly to the HMD control unit (for stereoscopic viewing). The control unit also receives stereo sound, which is then sent to the HMD built-in headphones. The user's head motion is tracked and position data sent back to the VR engine for use in the graphics computations. A ratchet on the back of the HMD head support allows comfort adjustment for various head sizes.

The HMD weight, comfort, and cost are additional criteria to be considered in comparing the various models on the market. Technology has made tremendous advances in miniaturizing the HMD displays and associated optics. The first LCD-based HMD (the VPL EyePhone), which was available in the early 1990s, had a resolution of 360×240 pixels, a FOV of 100° horizontally and 60° vertically, and a weight of 2.4 kg (2400 grams). This large weight induced user fatigue. Modern LCD-based

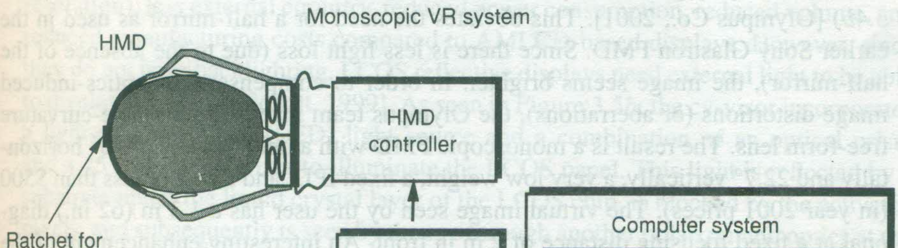


Fig. 3.3 Head-mounted display (HMD) integration in a VR system for: (*a*) consumer (monoscopic) HMD; (*b*) professional (stereoscopic) HMD. Adapted in part from Pimentel and Teixeira [1993]. Reprinted by permission.

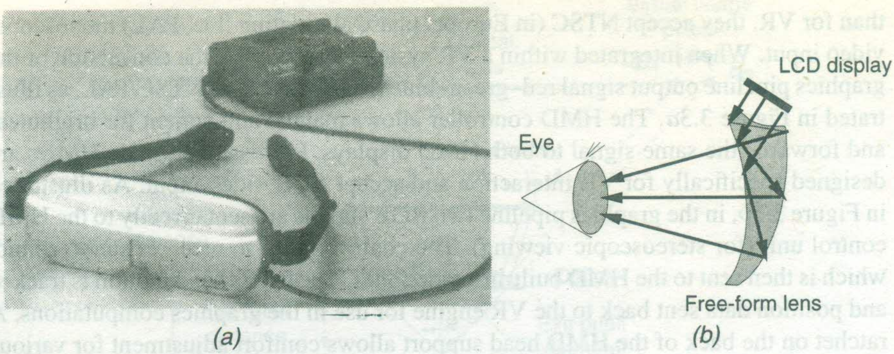


Fig. 3.4 The Olympus LCD-based Eye-Trek FMD: (*a*) the FMD 200; (*b*) optics arrangement. Adapted from Olympus Co. [2001]. Reprinted by permission.

HMDs, such as the Olympus Eye-Trek, shown in Figure 3.4*a* [Isdale, 2000], weighs only about 100 grams (24 times less!). This class of HMDs resembles eyeglasses, and so they are also called face-mounted displays (FMDs). Key to the compactness and lightness of the Olympus FMD is the placement of its small active matrix LCD (AMLCD) display panels eccentrically, directly above the optics (as shown in Figure 3.4*b*) [Olympus Co., 2001]. This obviates the need for a half-mirror as used in the earlier Sony Glastron FMD. Since there is less light loss (due to the absence of the half-mirror), the image seems brighter. In order to compensate for optics-induced image distortions (or aberrations), the Olympus team invented a variable-curvature free-form lens. The result is a monoscopic FMD with a field of view of 30° horizontally and 22.7° vertically, a very low weight, a fixed IPD, and a cost of less than $500 (in year 2001 prices). The virtual image seen by the user has a 1.3 m (62 in.) diagonal at a fixed focusing distance of 2 m in front. An interesting enhancement of the FMD 200, recently introduced in Europe, is the cordless TV. A pair of miniature TV transmitter and receiver units, operating in the 2.4-GHz audio/video frequency range, allows viewing up to 100 m from the TV. This greatly enhances the user's freedom of motion, at the expense of a larger system weight (the receiver is coupled with the FMD controller). Furthermore, interference from other home appliances using the same frequency range and from metal in the house construction can occur.

For all its ergonomic advantages, the Olympus FMD 200 has a number of drawbacks. It cannot be easily retrofitted with a head tracker, for which there are no attachments provided. A tracker receiver placed on top of the FMD will make the whole assembly fall off due to the added weight and the lack of a restraint at the back of the head. Another problem is the FMD 200 resolution, which is only 267 × 225 (or 180,000 pixels/LCD panel). Recognizing this problem, Olympus developed the FMD 700, which has the same FOV and four times the resolution of the FMD 200 model. The FMD 700 resolution of 533 × 450 (720,000 pixels/LCD panel) is obtained by adding special polarization and double-refraction effects, a process which Olympus calls optical super resolution.

Another LCD-based consumer FMD is the Daeyang cy-visor DH-4400VP shown in Figure 3.5*a*. Although it looks somewhat similar to the Eye-Trek FMD, the cy-visor uses liquid-crystal-on-silicon (LCOS) display technology. Classic AMLCD panels are transmissive, meaning that light passes through the electrode/liquid crystal/electrode

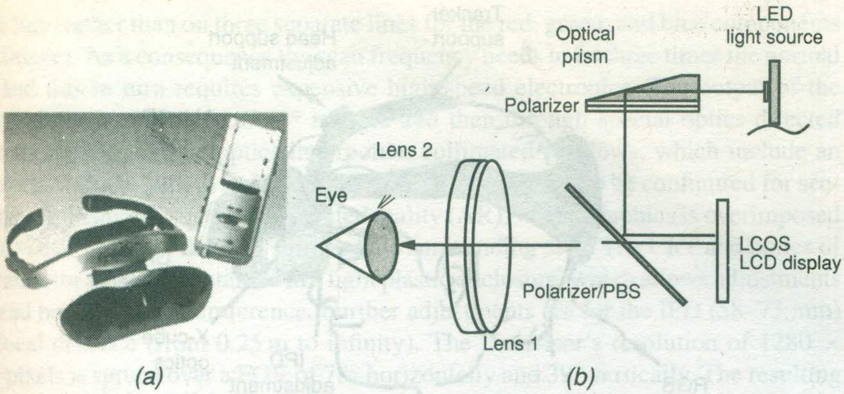


Fig. 3.5 The Daeyang cy-visor FMD: (*a*) the DH-4400VP; (*b*) optics arrangement. Adapted from Daeyang [2001]. Reprinted by permission.

substrate and is then seen by the user. The great advantage of newer LCOS LCD displays is that the liquid crystal layer is mounted directly on a silicon chip, which is used as control circuitry. This results in higher electrode densities (i.e., display resolution), less external circuitry, reduced power consumption, reduced volume, and reduced manufacturing costs compared to AMLCD-based displays. However, since they do not have backlighting, LCOS reflective displays need external light to be able to display the image [Schott, 1999]. As seen in Figure 3.5*b*, the cy-visor incorporates a light-emitting diode (LED) light source and a combination of an optical prism and a 45° mirror/polarizer to illuminate the LCOS panel. This light is reflected by a substrate under the liquid crystal layer of the LCOS chip, is blocked by the activated pixels, and subsequently is seen by the eye through another polarizer and optics at the front end of the FMD. By using this new technology the cy-visor is able to display an SVGA-resolution image (800 × 600 pixels) with a 60° horizontal × 43° vertical FOV. The use of the LCOS chips is one reason why the cy-visor is light (160 grams) while having an image resolution superior to any of the Eye-Trek FMD models previously described.

Even with improved resolution, consumer FMDs cannot match the professional-grade LCD-based HMDs. An example of such devices is the ProView XL35 illustrated in Figure 3.6*a* [Kaiser Electro-Optics, 2001]. It has high-resolution AMLCD displays located at the user forehead and attached to an ergonomically shaped adjustable head support. The support allows for the placement of a tracker close to the top of the head, while at the same time serving as a counterweight to reduce the user's fatigue. Each display has three separate LCD panels, with red, green, and blue backlights and an "X-cube" optical path for each eye. This color addition approach has fewer visual artifacts than single-panel based displays. This results in a 28° horizontally and 21° vertically stereo (100% overlap) FOV, comparable to that of the FMD 200. However, the ProView XL35 has a much higher XGA resolution (1024 × 768 × 3 pixels). The high resolution, combined with a relatively small FOV, results in a very low virtual image granularity (1.6 arc-minutes/color pixel). Further advantages are the adjustable IPD (55–75 mm) and an eye relief that allows eyeglasses to be worn together with the ProView HMD. The high image quality of the ProView XL35 comes at a price in

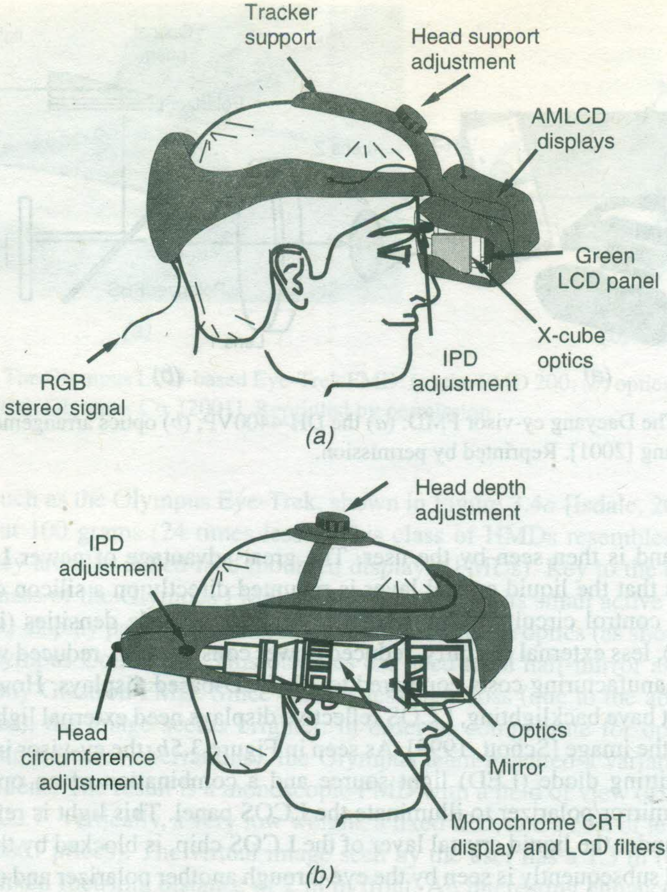


Fig. 3.6 Stereoscopic HMDs. (*a*) ProView XL35. Adapted from Kaiser Electro-Optics [2001]. Reprinted by permission. (*b*) Datavisor HiRes. Adapted from n-vision Inc. [1998].

terms of much higher cost ($19,500) and weight (992 grams, or 35 oz) compared to those of the FMD 200.

The high resolution of the ProView XL35 HMD represents only 60% of that available on desk-top graphics monitors, which typically have 1280 × 1024 color pixels. To obtain such display resolutions, today's high-end HMDs use miniature (1-in. diameter) CRTs. An example is the Datavisor HiRes, illustrated in Figure 3.6*b* [n-vision Inc., 1998]. Its two monochrome (black/white) CRTs are placed laterally to the user's head and retrofitted with small Tektronix Nucolor liquid crystal shutters. The CRTs and shutter units are metal-shielded to eliminate electronic interference and to dissipate heat away from the user's head. Special care is required in the electrical circuitry because of the high voltage of the CRTs, which are close to the user's head.

By rapidly switching the filters, a single line in the Datavisor's image is first seen in red, then in green, and finally in blue. The human brain integrates the three images to obtain a color scene. This approach in turn requires the so-called field-sequential video input format, meaning that RGB color information is transmitted sequentially on a single line (rather than on three separate lines for the red, green, and blue components of an image). As a consequence the scan frequency needs to be three times the normal rate, and this in turn requires expensive high-speed electronics. The output of the LCD shutters is reflected off 45° mirrors and then through special optics directed to the user's eyes. These optics incorporate collimated windows, which include an undistorted optical path to the eye. This allows the Datavisor to be configured for see-through applications, such as augmented reality (AR), where graphics is overimposed on the visual image of the user's immediate surrounding area. The CRT and optics of the Datavisor are fully contained in a light plastic enclosure, which allows adjustments for head height and circumference. Further adjustments are for the IPD (58–73 mm) and focal distance (from 0.25 m to infinity). The Datavisor's resolution of 1280 × 1024 pixels is spread over a FOV of 78° horizontally and 39° vertically. The resulting image granularity is 1.9 arc-minutes/color pixel, which is about 20% worse than that of the ProView XL35 LCD HMD. Other drawbacks of the Datavisor are large weight (1587 grams, or 56 oz) and unit cost ($35,000). This is further complicated by safety concerns related to the CRT electromagnetic radiation close to the brain.

the resolution drops by orders of magnitude. Increasing the resolution of very large displays to that of workstations would involve many more projectors and an even larger cost than those given in Table 3.2.

3.2 SOUND DISPLAYS

Definition Sound displays are computer interfaces that provide synthetic sound feedback to users interacting with the virtual world. The sound can be monoaural (both ears hear the same sound) or binaural (each ear hears a different sound).

Sound displays play an important role in increasing the simulation realism by complementing the visual feedback provided by the graphics displays previously discussed. Assume a user is looking at a virtual ball bouncing in a virtual room that is displayed on a CRT monitor. The user's mind says that he or she should also hear a familiar "plop-plop-plop" sound. When sound is added, the user's interactivity, immersion, and perceived image quality increase. In this scenario simple monoaural sound is sufficient, as the ball is always in front of the user and being displayed by the monitor. Let us now consider another user, who looks at the same virtual world, this time displayed by an HMD. If the ball bounces away, outside the field of view (as balls in the real world sometimes do), then the user cannot tell where the ball went based on visual information or on monoaural sound alone. Now the simulation needs a device that provides binaural sound in order to localize the "plop-plop-plop" sound in 3D space relative to the user.

This example illustrates an important distinction within the sound feedback modality. Highly immersive virtual reality simulations should have 3D sound, also called virtual sound, in addition to graphics feedback. The distinction between 3D sound and stereo sound is illustrated in Figure 3.20 [Burdea and Coiffet, 1993]. Stereo sound on headphones seems to emanate inside the user's head. In other words it is not externalized, as real sounds are. When the user wears simple stereo headphones, the violin will turn to the left following the user's head motion. A 3D sound presented on the same headphones or on speakers contains significant psychoacoustic information to alter the user's perception into believing that the recorded sound is actually coming from the user's surroundings [Currell, 1992]. In Figure 3.20, the 3D sound is synthesized using head tracker data, and the virtual violin remains localized in space. Therefore its sound will appear to move to the back of the user's head. Sound in a real room bounces off the walls, the floor, and the ceiling, adding to the direct sound received from the violin. The realism of the virtual room therefore requires that these reflected sounds be factored in.

3.2.1 The Human Auditory System

Three-dimensional sound displays cannot be effective without an understanding of the way we localize sound sources in space. Humans perceive sound through vibrations arriving at the brain via the skeletal system or via the ear canal. Within the scope of this book we are interested in the ear's ability to detect the position of a sound source relative to the head.

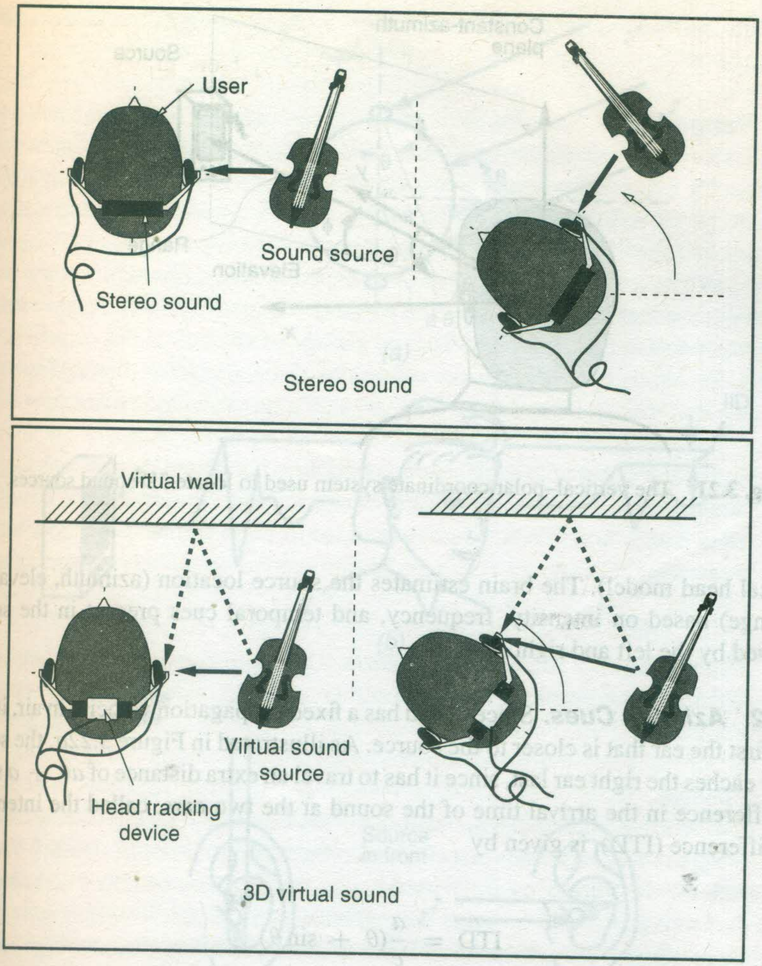


Fig. 3.20 Stereo sound versus 3D virtual sound. From Burdea and Coiffet [1993]. © Editions Hermes. Reprinted by permission.

3.2.1.1 The Vertical–Polar Coordinate System. In order to measure position it is necessary to establish a head-attached system of coordinates. If such a system of coordinates is Cartesian, then the location of a sound source relative to the head can be expressed by the triad (x, y, z). Alternatively, the same 3D sound source position can be expressed in a spherical system of coordinates called the vertical–polar coordinate system [Duda, 1997]. As illustrated in Figure 3.21, the sound source location is uniquely determined by three variables, namely azimuth, elevation, and range. The azimuth is the angle θ between the nose and a plane containing the source and the vertical axis z. The source elevation is the angle ϕ made with the horizontal plane by a line passing through the source and the center of the head. Finally, the range r is the distance to the source measured along this line. The sound source azimuth can vary anywhere between $\pm 180°$, while the elevation has a range of $\pm 90°$. Finally, source range can only be positive and larger than the head radius (assuming a simplified

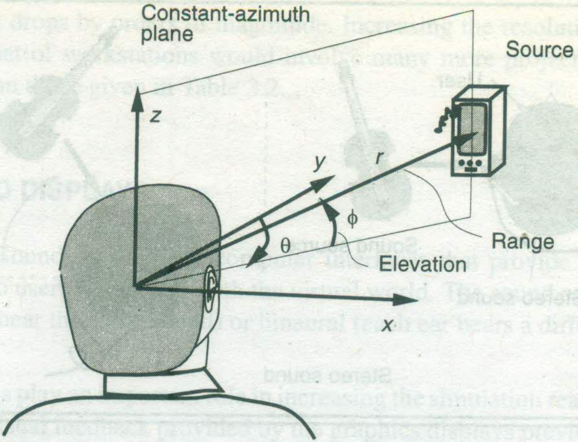


Fig. 3.21 The vertical–polar coordinate system used to locate 3D sound sources.

spherical head model). The brain estimates the source location (azimuth, elevation, and range) based on intensity, frequency, and temporal cues present in the sound perceived by the left and right ears.

3.3 HAPTIC FEEDBACK

The last category of I/O devices discussed in this chapter are haptic interfaces. Named after the Greek term *hapthai* (meaning "touch"), they convey important sensorial information that helps users achieve tactile identification of virtual objects in the environment and move these objects to perform a task [Cutt, 1993; Burdea, 1996]. When added to the visual and 3D audio feedback previously discussed, haptic feedback greatly improves simulation realism. It becomes mandatory wherever the visual feedback is corrupted (occluded objects that need to be manipulated) or lacking entirely (dark environments). The discussion in this chapter is limited to feedback to the user's hand and wrist because the hand has the highest density of touch receptors in the body. Furthermore, systems that provide feedback to the whole body are either in the research stage at this time or are too specialized to be of use outside their intended settings.

Haptic feedback groups the two modalities of touch and force feedback. Consider a hand pushing very lightly against a desk. The sensors that respond first are the fingertip touch sensors. If the hand pushes harder, then the muscles in the hand and forearm start to contract. The level of force applied is now felt by sensors on muscle ligaments and bones (kinesthesis) as opposed to fingertips. The technical literature intermixes the terms "touch" and "force" feedback, so clarification is needed.

Definition Touch feedback conveys real-time information on contact surface geometry, virtual object surface roughness, slippage, and temperature. It does not actively resist the user's contact motion and cannot stop the user from moving through virtual surfaces.

Definition Force feedback provides real-time information on virtual object surface compliance, object weight, and inertia. It actively resists the user's contact motion and can stop it (for large feedback forces).

Designing good haptic feedback interfaces is a daunting task in view of the specific requirements they need to meet. The first is user safety and comfort. While the user interacts with virtual objects, the forces he or she feels are *real*. These contact forces need to be large (in the simulation of rigid objects), but not large enough to harm the user. In this context a good design is also fail-safe, so that users are not subject to accidents in case of computer failure. Yet another issue is portability and user comfort. The difficulty with force-feedback actuators is the need to provide sufficient force while still keeping the feedback hardware light and unintrusive. If haptic interfaces are too heavy and bulky, then the user will get tired easily and will prefer a less cumbersome open-loop control. Heavy feedback structures can be gravity counterbalanced, but this further increases complexity and cost. Portability also relates to ease of use and installation at the simulation site. Haptic feedback interfaces should be self-contained, without requiring special supporting construction, piping, or wiring.

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All these conflicting requirements call for a compromise, which can only be found if the human haptic characteristics are taken into account. In the context of this book we look at the sensorial characteristics of the hand from the engineering perspective. The hardware designer is interested in the manual force exertion capability, which helps in selecting the appropriate feedback actuators. The hardware designer is also interested in manual degrees of freedom and mobility, which determine the number of feedback actuators and their range of motion. Similarly, the control engineer is interested in the sensorial bandwidth of the hand, which determines how fast the control loop needs to be closed. Finally, the human factors specialist is interested in the safety and comfort issues related to feedback to the hand.

3.3.1 The Human Haptic System

Input to the human haptic system is provided by the sensing loop, while output to the environment (in this case the haptic interface) is mediated by the sensory–motor control loop. The input data are gathered by a multitude of tactile, proprioceptive, and thermal sensors, while the output is forces and torques resulting from muscular exertion. The system is not balanced, in the sense that humans perceive hapticly much faster than they can respond.

3.3.1.1 Haptic Sensing. The skin houses four types of tactile sensors (or receptors), namely Meissner corpuscles (the majority), Merkel disks, Pacinian corpuscles, and Ruffini corpuscles. When excited they produce small electrical discharges, which are eventually sensed by the brain. The slow-adapting (SA) sensors (Merkel and Ruffini) maintain a constant rate of discharge for a long time in response to a constant force being applied on the skin. On the contrary, the fast-adapting (FA) ones (Meissner and Pacinian) drop their rate of discharge so fast that the constant contact force becomes undetected. Such different behavior is related to the particular type of contact the tactile sensors detect. If the contact has a strong time variation (vibrations on the skin, or accelerations), then FA receptors are needed. Conversely, if the contact is quasi steady state (edges, skin stretch, static force), then SA receptors are used. As a consequence, Merkel and Ruffini receptors are most sensitive to low-frequency stimuli (0–10 Hz), while Meissner and Pacinian sensors respond to stimuli of higher frequencies (50–300 Hz). The receptive field of a tactile receptor determines its spatial resolution. If the sensor has a large receptive filed, its spatial resolution is low. Conversely, the Meissner and Merkel receptors, which have small receptive fields, have high spatial resolution. Table 3.3 summarizes the characteristics of the four tactile sensors (or receptors) present in the skin [Burdea, 1996].

The spatial resolution of the skin varies depending on the density of receptors it has. The fingertips, where such density is the highest, can discriminate two contacts that are at least 2.5 mm apart. However, the palm cannot discriminate two points that are less than 11 mm apart, and the user feels as if only one contact point exists [Shimoga, 1992]. Spatial resolution is complemented by temporal resolution when two contacts occur on the skin close in time. The skin mechanoreceptors have a successiveness threshold of only 5 msec, which is much smaller than the corresponding value for the eye (25 msec).

Temperature sensing is realized by specialized thermoreceptors and nociceptors. The thermoreceptors are located either in the epidermis (sensitive to cold) or dermis (sensitive to warmth). Their spatial resolution is less than that of mechanoreceptors.

TABLE 3.3. Comparison of Various Skin Mechanoreceptors[a]

Receptor Type	Rate of Adaptation[b]	Stimulus Frequency (Hz)	Receptive Field	Detection Function
Merkel disks	SA-I	0–10	Small, well defined	Edges, intensity
Ruffini corpuscles	SA-II	0–10	Large, indistinct	Static force, skin stretch
Meissner corpuscles	FA-I	20–50	Small, well defined	Velocity, edges
Pacinian corpuscles	FA-II	100–300	Large, indistinct	Acceleration, vibration

[a] Based on Seow [1988], Cholewiak and Collins [1991], and Kalawsky [1993].
[b] SA, slow-adapting; FA, fast-adapting.